



Logistics Optimization for Delivery Routes – FedEx

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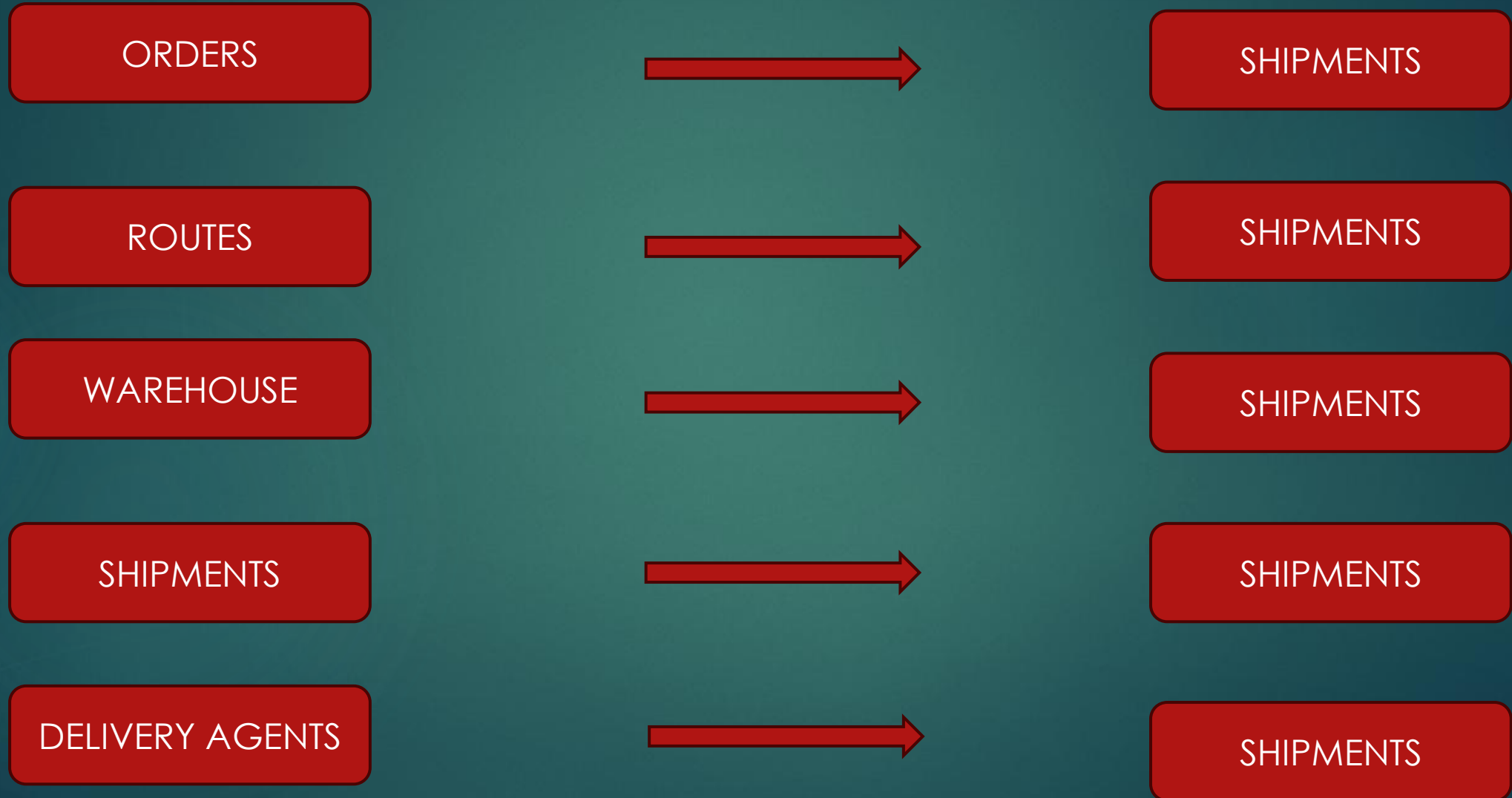
COURSE: SQL PROJECT

DATE: JUNE 2026

Project Objective

- ▶ • Analyze shipment delays
- ▶ • Optimize delivery routes
- ▶ • Evaluate warehouse performance
- ▶ • Evaluate delivery agent performance
- ▶ • Generate KPI reports
- ▶ • Recommend operational improvements

Database Schema



Task 1 – Data Cleaning & Preparation

► Query Answer

```
51 -- Task 1 : Data Cleaning & preparation
52
53
54 • SELECT Order_ID, COUNT(*)
55 FROM Orders
56 GROUP BY Order_ID
57 HAVING COUNT(*) > 1;
58
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Order_ID	COUNT(*)
----------	----------

```
66 • SELECT Shipment_ID, COUNT(*)
67 FROM Shipments
68 GROUP BY Shipment_ID
69 HAVING COUNT(*) > 1;
70
71 • UPDATE Shipments s
72 JOIN (
73 SELECT Route_ID,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Shipment_ID	COUNT(*)
-------------	----------

```
94 • SELECT *
95 FROM Shipments
96 WHERE Delivery_Date < Pickup_Date;
97
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Shipment_ID	Order_ID	Agent_ID	Route_ID	Warehouse_ID	Pickup_Date	Delivery_Date	Delivery_Status	Delay_Hours	Delivery_Feedback	Delay_Reason
-------------	----------	----------	----------	--------------	-------------	---------------	-----------------	-------------	-------------------	--------------

```
98 • SELECT 'Shipment-Order' AS Check_Type,
99 COUNT(*) AS Invalid_Records
100 FROM Shipments s
101 LEFT JOIN Orders o
102 ON s.Order_ID = o.Order_ID
103 WHERE o.Order_ID IS NULL
104
105 UNION ALL
106
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Check_Type	Invalid_Records
Shipment-Order	0
Shipment-Route	0
Shipment-Warehouse	0
Order-Route	0
Order-Warehouse	0

► FINDINGS

- ✓ No duplicate Shipment_ID records
- ✓ No duplicate Order_ID records
- ✓ Missing values handled
- ✓ Referential integrity validated
- ✓ No invalid delivery dates found

Task 2 – Delivery Delay Analysis

► Subtask 1

- Calculate delivery delay
- This calculates the actual transit time for each shipment.

► Subtask 2

- Top 10 delayed routes

```
150 #2)
151 • SELECT
152   Route_ID,
153   round(AVG(Delay_Hours),2) AS Avg_Delay
154 FROM Shipments
155 GROUP BY Route_ID
156 ORDER BY Avg_Delay DESC
157 LIMIT 10;
```

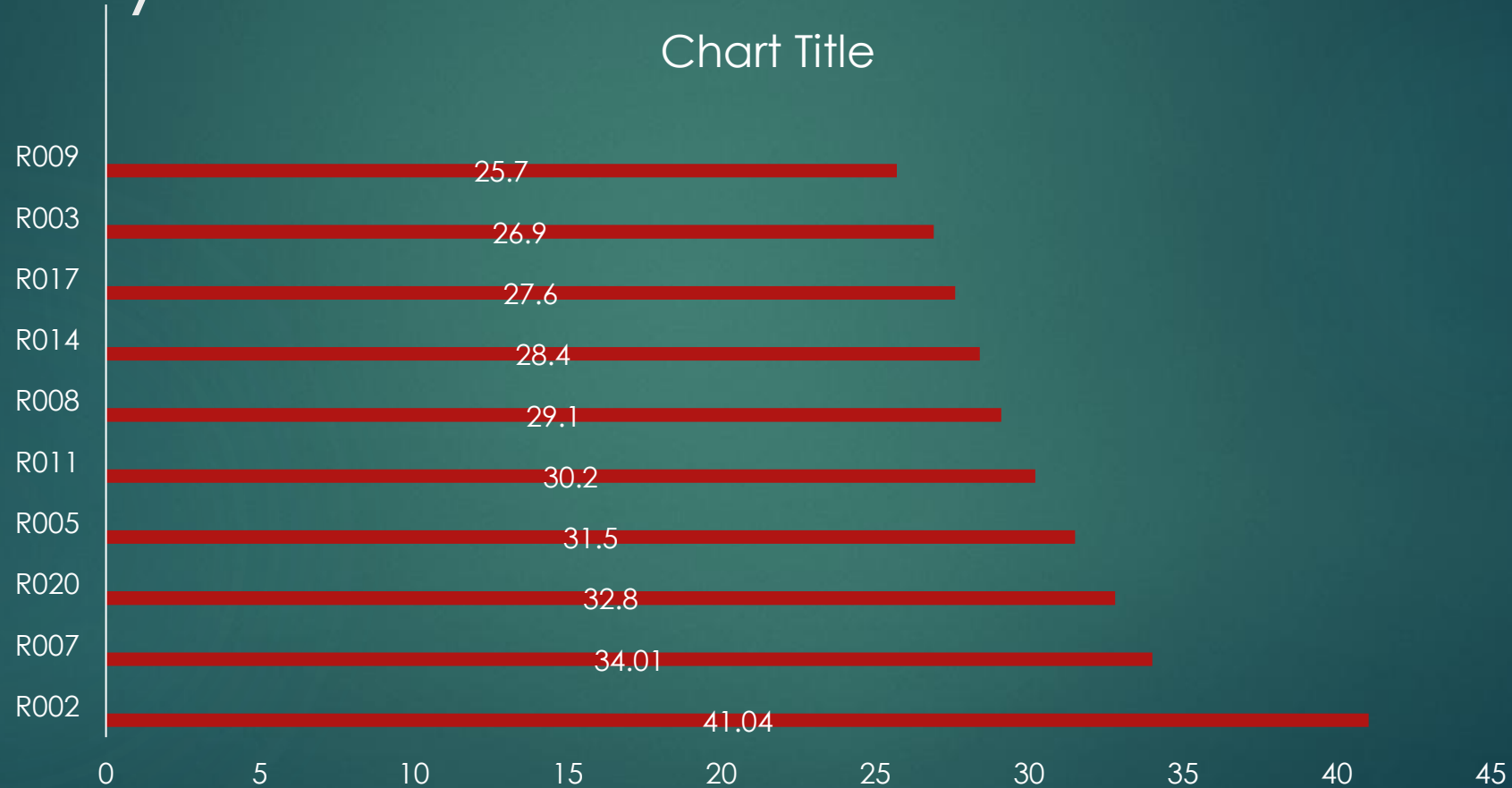
Route_ID	Avg_Delay
R002	41.04
R007	34.01
R020	32.8
R003	32.18
R008	30.81
R005	29.86
R004	25.88
R018	25.83
R011	22.64
R014	20.97

```
143 #1)
144 • select
145   shipment_id,
146   timestampdiff(hour,pickup_date, delivery_date) as Transit_hours
147 from shipments;
148
```

shipment_id	Transit_hours
S00001	22
S00002	26
S00003	21
S00004	26
S00005	16
S00006	20
S00007	15
S00008	21
S00009	99
S00010	25

R002	41.04
R007	34.01
R020	32.8
R003	32.18
R008	30.81
R005	29.86
R004	25.88
R018	25.83
R011	22.64
R014	20.97

Top 10 Delayed Routes by Average Delay Hours



TASK 2

SUB TASK 3

Ranks shipments by delay within each warehouse.

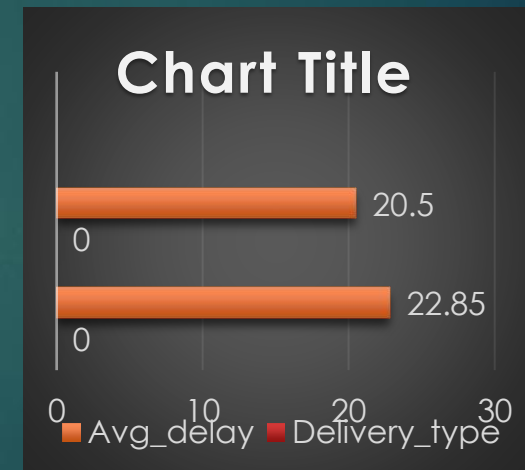
```
160 • SELECT
161   Shipment_ID,
162   Warehouse_ID,
163   Delay_Hours,
164   RANK() OVER(
165     PARTITION BY Warehouse_ID
166     ORDER BY Delay_Hours DESC
167   ) AS Delay_Rank
168 FROM Shipments;
169
```

Shipment_ID	Warehouse_ID	Delay_Hours	Delay_Rank
S00648	W001	205.7	1
S00670	W001	194.6	2
S00997	W001	193.7	3
S00349	W001	171.6	4
S00069	W001	170.8	5
S00148	W001	138.7	6
S00192	W001	99.5	7
S00153	W001	95.4	8
S00739	W001	71.4	9
S00816	W001	66.3	10
S00550	W001	64.5	11
S00193	W001	61.9	12

- ▶ Subtask 4:
- ▶ Average delay by Delivery Type

```
71 • SELECT
72   o.Delivery_Type,
73   ROUND(AVG(s.Delay_Hours),2) AS Avg_Delay
74 FROM Orders o
75 JOIN Shipments s
76 ON o.Order_ID = s.Order_ID
77 GROUP BY o.Delivery_Type;
78
```

Delivery_Type	Avg_Delay
Standard	20.85
Express	22.5



Task 3 – Route Optimization Analysis

Purpose:

Calculate average transit time for each route.

Business Impact:

Routes with high transit times may indicate congestion, poor planning, or operational inefficiencies.

```
2 SELECT
3   Route_ID,
4   ROUND(
5     AVG(TIMESTAMPDIFF(HOUR,Pickup_Date,Delivery_Date)),2
6   ) AS Avg_Transit_Time
7 FROM Shipments
8 GROUP BY Route_ID;
```

- ▶ Query Result
- ▶ Observation:
- ▶ Route R002 takes the longest average transit time.

Result Grid			Filter Rows:
	Route_ID	Avg_Transit_Time	
▶	R019	30.73	
	R011	41.13	
	R002	63.70	
	R020	52.98	
	R015	28.11	
	R008	51.14	
	R012	63.80	
	R009	28.18	
	R010	33.58	
	R018	75.73	
	R014	52.36	
	R016	51.63	

Route Efficiency Comparison

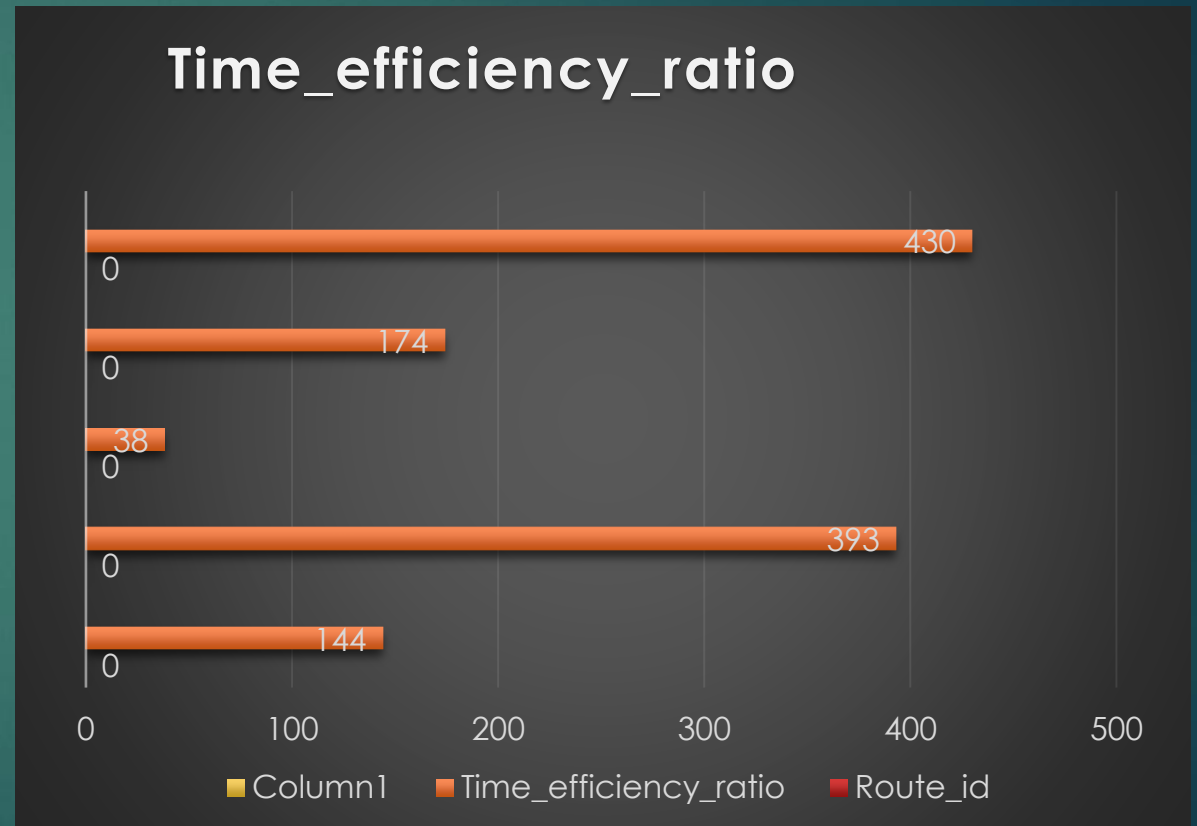
Insight:

Routes with lower efficiency ratios require optimization.
Route R004 shows the lowest efficiency ratio and should be investigated for delays, congestion, or routing issues.

#3)

```
select route_id, distance_km , avg_transit_time_hours,  
round( distance_KM/avg_transit_time_hours) as Time_efficiency_Ratio  
from routes;
```

Result Grid	Filter Rows:	Export:	Wrap Cell
route_id	distance_km	avg_transit_time_hours	Time_efficiency_Ratio
R001	4620	32.1	144
R002	7264	18.5	393
R003	885	23.5	38
R004	8555	49.1	174
R005	10023	23.3	430
R006	2364	23.3	101
R007	5297	30.5	174
R008	1846	17.1	108
R009	5582	15.5	360
R010	6268	11.6	540
R011	7369	15.5	475
R012	11397	42.3	269

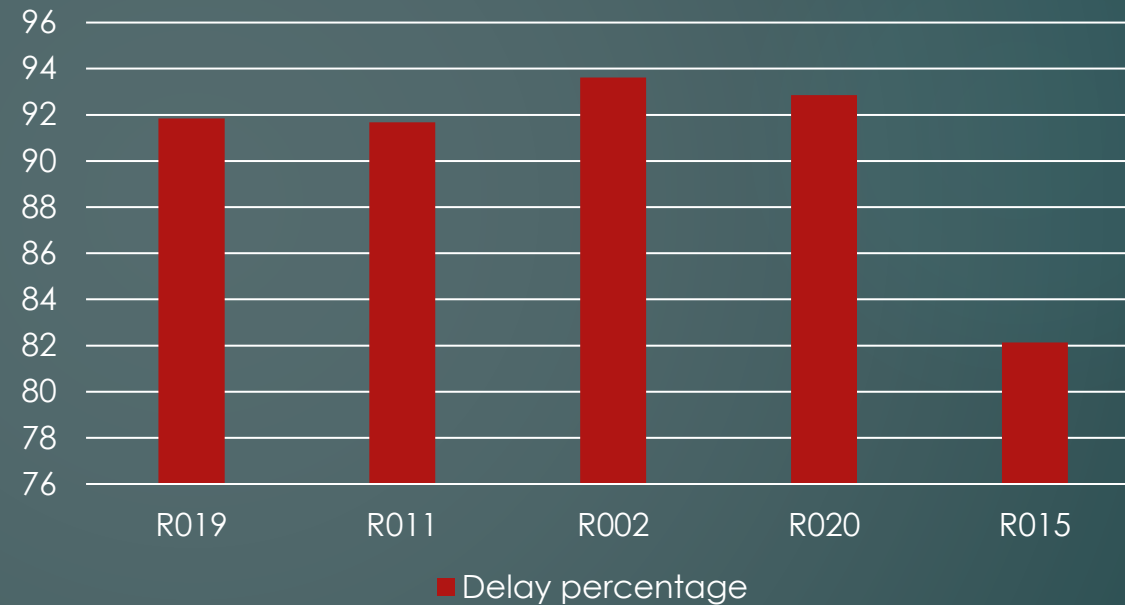


Task 3.4 – High Delay Routes (>20%)

Result Grid			Filter Rows:
	Route_ID	Delay_Percentage	
▶	R019	91.84	
	R011	91.67	
	R002	93.62	
	R020	92.86	
	R015	82.14	
	R008	90.91	
	R012	98.21	
	R009	100.00	
	R010	98.08	
	R018	94.92	
	R014	92.45	
	R016	88.73	

```
9 • SELECT
10 Route_ID,
11 ROUND(
12 SUM(CASE WHEN Delay_Hours > 0 THEN 1 ELSE 0 END)
13 *100.0/COUNT(*),2
14 ) AS Delay_Percentage
15 FROM Shipments
16 GROUP BY Route_ID
17 HAVING Delay_Percentage > 20;
```

Delay percentage

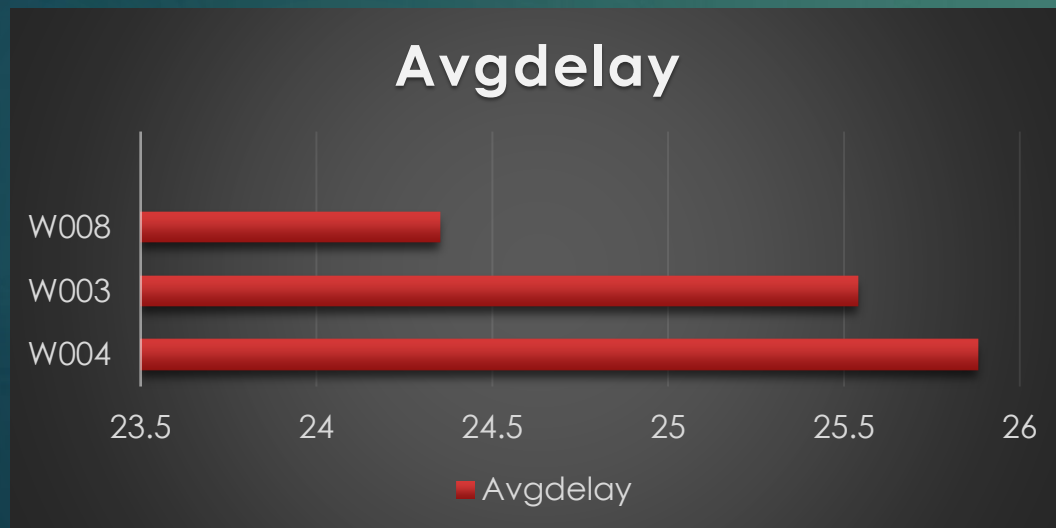


Task 4 Warehouse Performance

Task 4.1 – Top 3 Warehouses by Average Delay

```
select
warehouse_id ,
round(avg(delay_hours),2) as Avgdelay
from shipments
group by warehouse_id
order by avgdelay desc limit 3;
```

	warehouse_id	Avgdelay
▶	W004	25.88
	W003	25.54
	W008	24.35



Task 4.2 – Total vs Delayed Shipments

```
SELECT
Warehouse_ID,
COUNT(*) AS Total_Shipments,
SUM(
CASE
WHEN Delay_Hours > 0 THEN 1
ELSE 0
END
) AS Delayed_Shipments
FROM Shipments
GROUP BY Warehouse_ID;
```

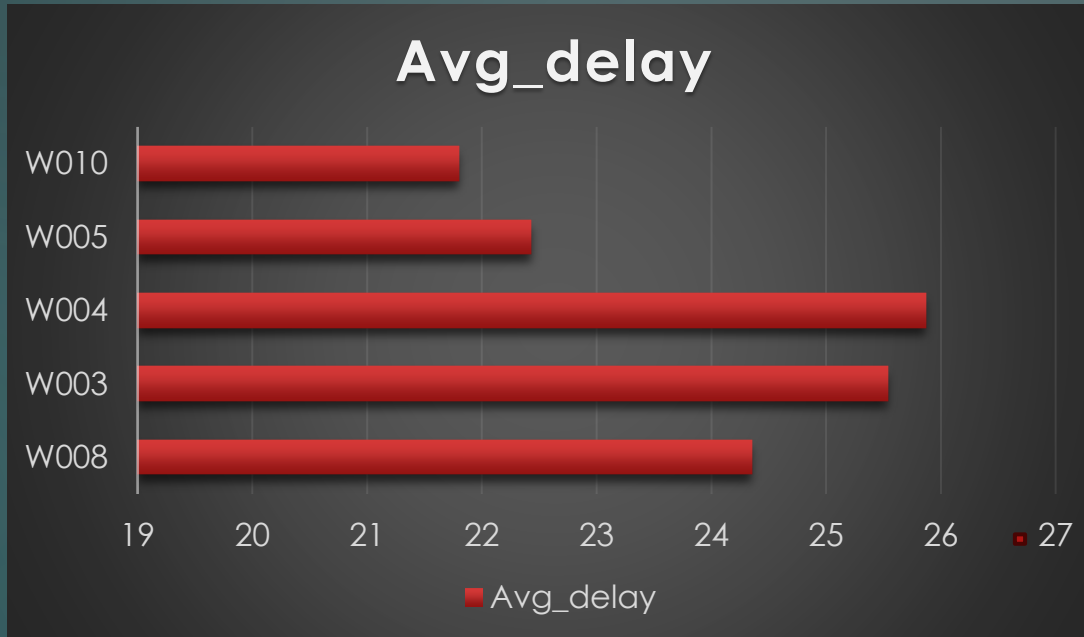
	Warehouse_ID	Total_Shipments	Delayed_Shipments
▶	W008	112	100
	W003	107	100
	W001	103	96
	W009	91	84
	W004	90	85
	W002	97	91
	W005	102	94
	W010	103	94
	W007	96	94
	W006	99	94



Task 4.3 Warehouses Above Global Average Delay

Global Average Delay =
18.5 Hours

INSIGHT:
These warehouses perform
worse than the network
average.



```
WITH GlobalAvg AS  
(  
    SELECT AVG(Delay_Hours) AS AvgDelay  
    FROM Shipments  
)  
SELECT  
    s.Warehouse_ID,  
    AVG(s.Delay_Hours) AS Avg_Delay  
FROM Shipments s  
CROSS JOIN GlobalAvg g  
GROUP BY s.Warehouse_ID  
HAVING AVG(s.Delay_Hours) > MAX(g.AvgDelay);
```

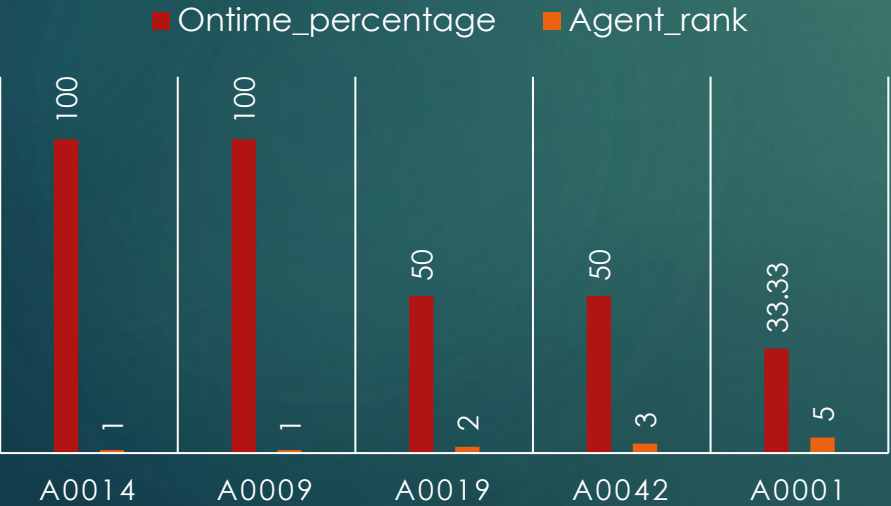
Result Grid		Filter Rows:
	Warehouse_ID	Avg_Delay
▶	W008	24.352678571428566
	W003	25.543925233644853
	W004	25.879999999999992
	W005	22.434313725490192
	W010	21.803883495145623
	W007	22.061458333333338

Task 5 Delivery Agent Performance Analysis,

Task 5.1 Agents by On-Time %

```
SELECT
  Agent_ID,
  Route_ID,
  ROUND(
    SUM(
      CASE
        WHEN Delay_Hours = 0 THEN 1
        ELSE 0
      END
    ) * 100.0 / COUNT(*), 2
  ) AS OnTime_Percentage,
  RANK() OVER(
    PARTITION BY Route_ID
    ORDER BY
      SUM(
        CASE
          WHEN Delay_Hours = 0 THEN 1
          ELSE 0
        END
      ) * 100.0 / COUNT(*) DESC
  ) AS Agent_Rank
FROM Shipments
GROUP BY Agent_ID, Route_ID;
```

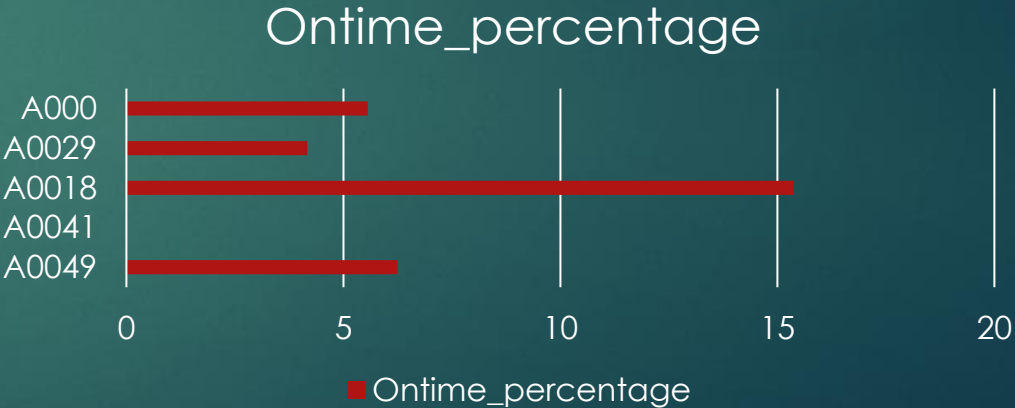
Agent_ID	Route_ID	OnTime_Percentage	Agent_Rank
A0014	R001	100.00	1
A0009	R001	100.00	1
A0019	R001	50.00	2
A0042	R001	50.00	3
A0001	R001	33.33	5
A0038	R001	0.00	6
A0033	R001	0.00	6
A0008	R001	0.00	6
A0025	R001	0.00	6
A0018	R001	0.00	6
A0005	R001	0.00	6
A0021	R001	0.00	6



Task 5.2 Average Delay by Agent

```
#2)
SELECT
  Agent_ID,
  ROUND(
    SUM(
      CASE
        WHEN Delay_Hours = 0 THEN 1
        ELSE 0
      END
    ) * 100.0 / COUNT(*), 2
  ) AS OnTime_Percentage
FROM Shipments
GROUP BY Agent_ID
HAVING OnTime_Percentage < 85;
```

Agent_ID	OnTime_Percentage
A0049	6.25
A0041	0.00
A0018	15.38
A0029	4.17
A0040	5.56
A0004	4.76
A0003	0.00
A0024	5.00
A0019	21.74
A0023	6.90
A0020	0.00
A0006	0.00



Task 6 Shipment Tracking Analytics

Task 6.1 Shipment Status Distribution

```
select
Shipment_ID,
Delivery_Status,
MAX(Delivery_Date) AS Latest_Delivery_Date
FROM Shipments
GROUP BY Shipment_ID, Delivery_Status;
```

	Shipment_ID	Delivery_Status
▶	S00001	Delivered
	S00002	Delivered
	S00003	Delivered
	S00004	Delivered
	S00005	Delivered
	S00006	Delivered
	S00007	Delivered
	S00008	Returned
	S00009	Delivered
	S00010	Delivered
	S00011	Delivered
	S00012	In Transit

Most shipments were successfully delivered, while a small percentage remain in transit or were returned.

Task 6.2 Routes with Majority In Transit / Returned Shipments

```
SELECT
Route_ID,
Delivery_Status,
COUNT(*) AS Shipment_Count
FROM Shipments
WHERE Delivery_Status IN ('In Transit','Returned')
GROUP BY Route_ID, Delivery_Status
ORDER BY Shipment_Count DESC;
```

	Route_ID	Delivery_Status	Shipment_Count
▶	R018	In Transit	11
	R017	In Transit	7
	R002	In Transit	6
	R016	In Transit	6
	R020	In Transit	6
	R012	In Transit	6
	R015	In Transit	5
	R013	In Transit	5
	R010	In Transit	4
	R004	In Transit	4
	R007	In Transit	4
	R013	Returned	4

Route R002 has the highest concentration of unresolved shipments.

Task 6

```
SELECT
Order_ID,
Shipment_ID,
Delay_Hours
FROM Shipments
WHERE Delay_Hours > 120;
```

	Order_ID	Shipment_ID	Delay_Hours
▶	O0093	S00015	238.6
	O0072	S00029	151.6
	O0214	S00035	213.1
	O0181	S00039	197.3
	O0197	S00052	230.1
	O0070	S00057	122.2
	O0173	S00069	170.8
	O0245	S00096	210.9
	O0203	S00101	218
	O0274	S00108	152.1
	O0233	S00120	180.6
	O0107	S00148	138.7

These orders represent extreme delays and should be investigated for bottlenecks.

Task 7 – Logistics KPI Dashboard

Result Grid	
Total_Shipments	
▶	1000

Result Grid	
Avg_Delay	
▶	21.41

Result Grid	
Total_Warehouses	
▶	10

Result Grid	
Total_Routes	
▶	20

Result Grid	
Total_Agents	
▶	50

Result Grid	
OnTime_Percentage	
▶	6.80

